CS & IT ENGINEERING

Operating System

Process Synchronization

Lecture - 01

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Recap of Previous Lecture

Topic System Call: Fork()

Topic Multithreading

Topics to be Covered

Topic Synchronization

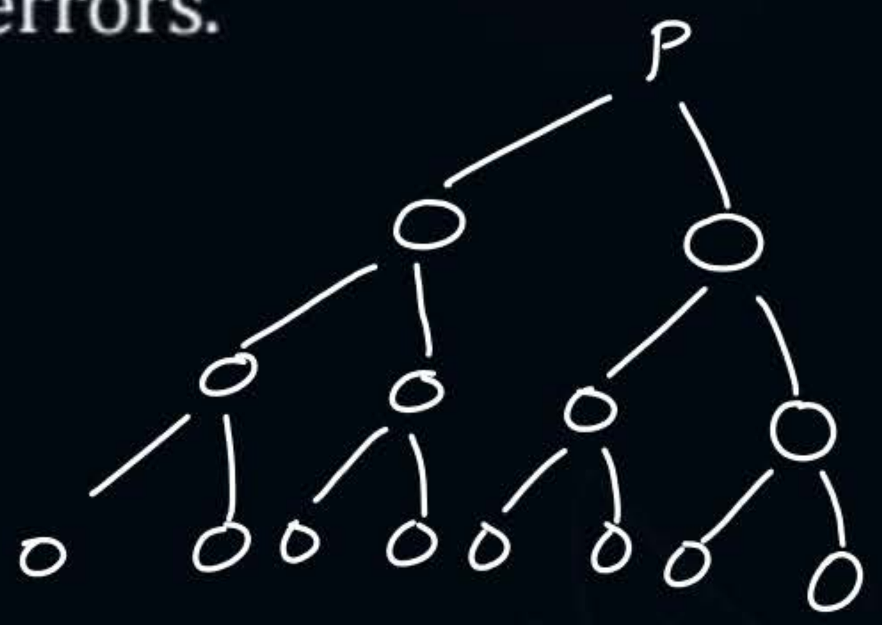
Topic Race Condition

Topic Critical Section

[NAT]

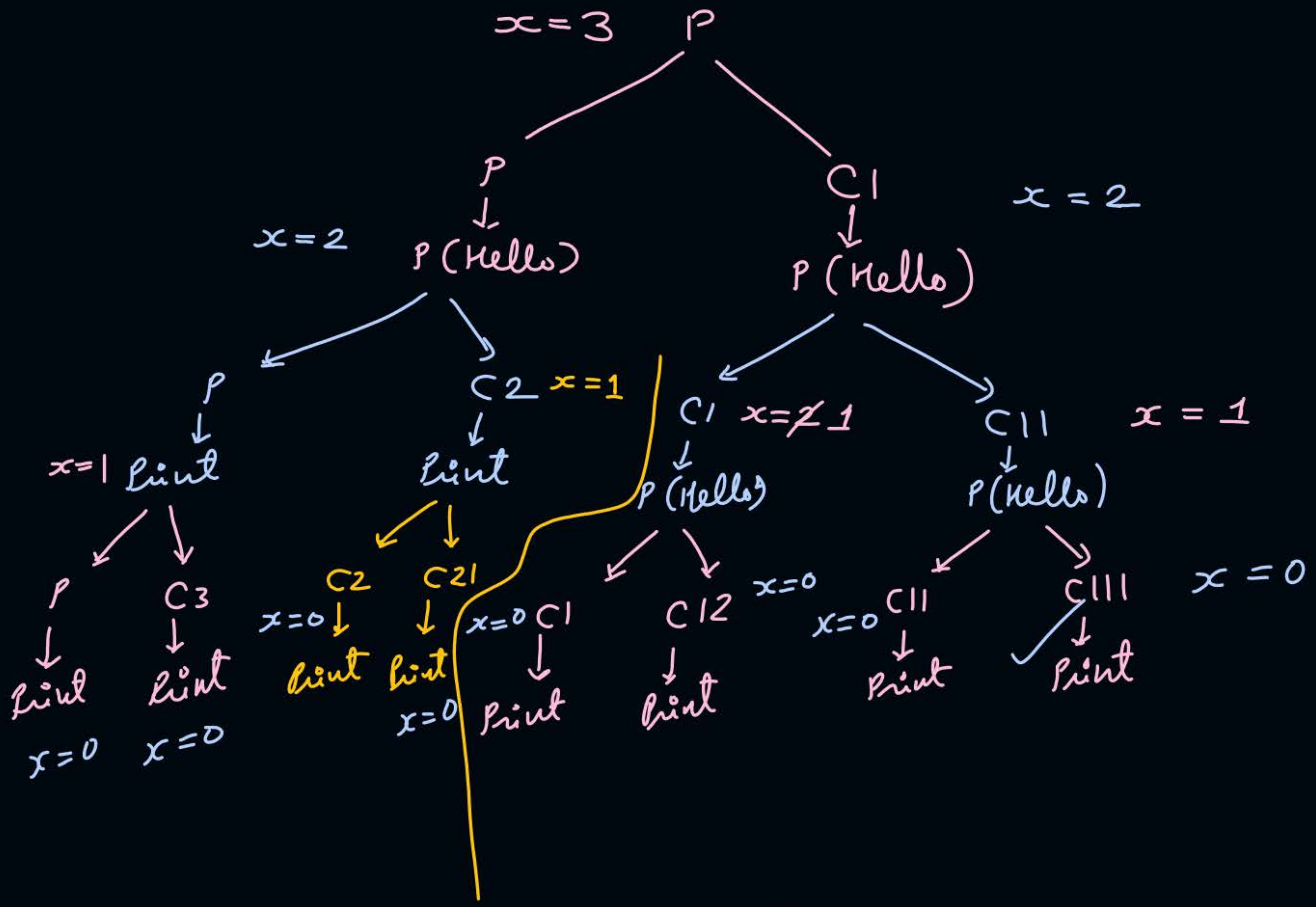
#Q. Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
fork();
printf("hello");
wait(NULL);
x--;
}
```



The total number of times the printf statement is executed is 14 ?

GATE-2024



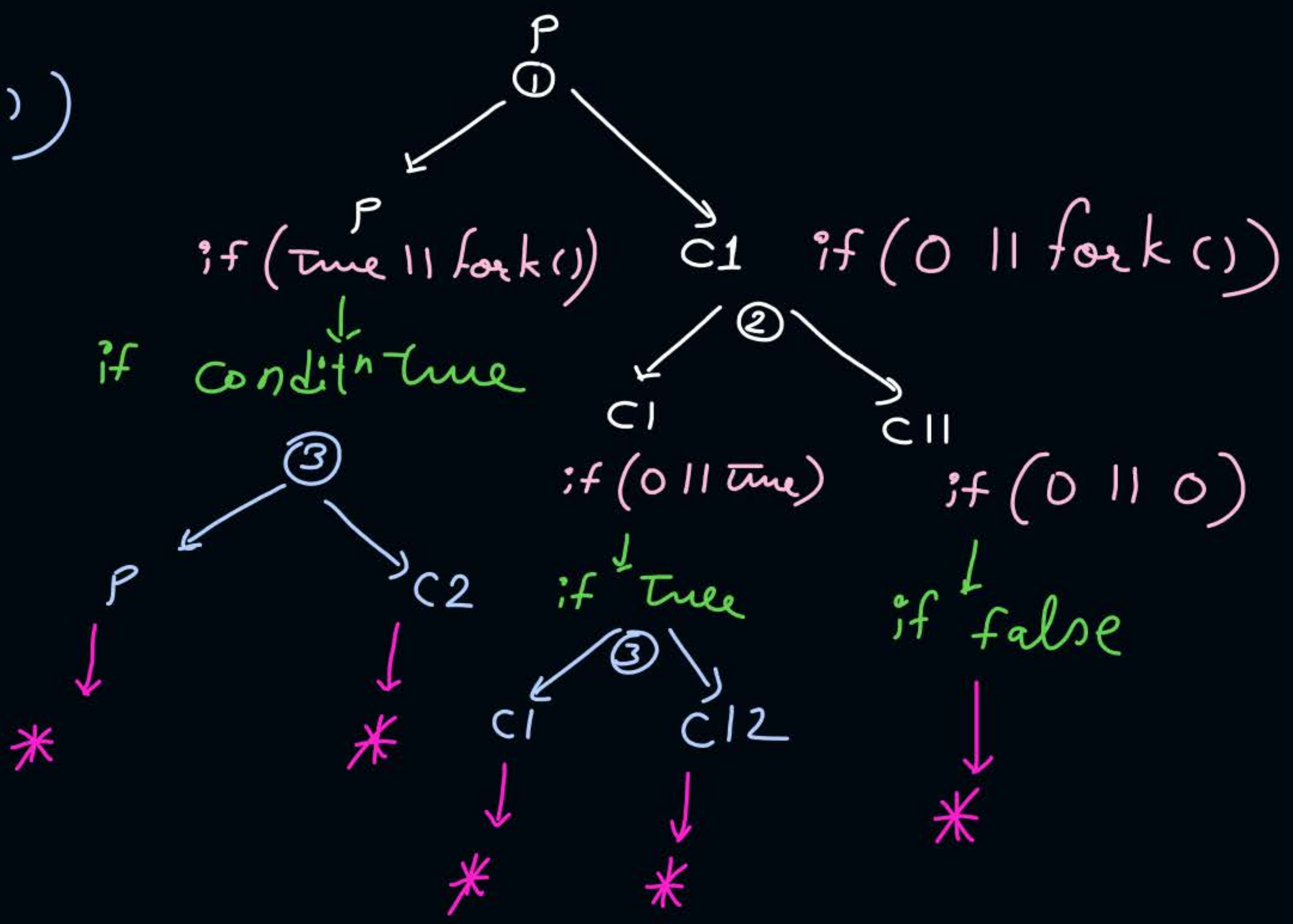
x=3 P
P
C1
x = 2
x=2
P (Hello)
P (Hello)
P
C2 x=1
C1 x=~~2~~1
C11
x = 1
x=1 Print
Print
P(Hello)
P(Hello)
P
C3
C2
C21
x=0
C12
x=0 C11
C111
x = 0
x=0 C1
Print
x=0

```
if (fork() || fork())
{
    fork();
}

print("*");
```

(1, 2, 3 written above the first, second and third fork)

Ans = 5



Practice:-

```
void main()
{
  if(fork() && fork())
  {
    fork();
  }
  fork();
  printf("*");
}
```

no. of times '*' is printed = ____ ?

Topic : Process Types

1. Independent → Processes which do not communicate with other processes

2. Cooperating/Coordinating/Communicating

   ↓

   communicate with other processes

Topic : Need Of Synchronization

communicating processes must execute in such way that the final result generated should be as per the expectatn.

Topic : Problems Without Synchronization

Problems without Synchronization:

- Inconsistency
- Loss of Data
- Deadlock

Topic : Entire Process Requires Synchronization?

$x = 5$

Process

Inst^{ns}

Remainder section

Communication (shared variable, message passing)

critical section

These inst^{ns} execution will need synchronizat^{n}

remainder section

Topic : Critical Section

The critical section is a code segment where the shared variables can be accessed.

Topic : Race Condition

A race condition is an undesirable situation, it occurs when the final result of concurrent processes depends on the sequence in which the processes complete their execution.

$x = 5$

P1	P2
$R1 = x$	$R2 = x$
$R1 = R1 + 2$	$R2 = R2 + 3$
$x = R1$	$x = R2$

P1	P2	P1	P2	P1	P2
$R1 = x$ $= 5$	$R2 = x$ $= 5$	$R1 = R1 + 2$ $= 7$	$R2 = R2 + 3$ $= 8$	$x = R1$ 7	$x = R2$ 8

$$x = \cancel{5}\ \cancel{7}\ 8$$

P2	P1	P2	P1	P2	P1
$R2 = x$ 5	$R1 = x$ $= 5$	$R2 = R2 + 3$ 8	$R1 = R1 + 2$ $= 7$	$x = R2$ 8	$x = R1$ 7

$$x = \cancel{5}\ \cancel{8}\ 7$$

[NAT]

#Q. X = 10

$R1 \leftarrow x$
$R1 \leftarrow R1/2$
$x \leftarrow R1$

P1

X = X /2
5 10

P2

X = X+4
14 10

How many different values of X are possible after both processes finish executing?

Ans = 4

Case 1 :-

Run P1 first completely, then P2 next

$x = \cancel{10}\ \cancel{5}\ 9$

Case 2 :-

Run P2 completely then P1

$x = \cancel{10}\ \cancel{14}\ 7$

Case 3 :-

Both read x together & P1 finishes last

$x = \cancel{10}\ \cancel{14}\ 5$

Case 4 :-

Both read x together & P2 finishes last

$x = \cancel{10}\ \cancel{5}\ 14$

[NAT]

#Q. The following pair of processes share a common variable X.

Process A	Process B
int Y;	int Z;
Y = X*2; 8	Z = X+1; 5
X = Y;	X = Z;

$x = 4$

1) $x = \cancel{8}$ 9

2) $x = \cancel{4}\cancel{8}$ 10

3) $x = \cancel{4}$ 8

4) $x = \cancel{4}$ 5

X is set to 4 before either process begins execution. As usual, statements within a process are executed sequentially, but statements in process A may execute in any order with respect to statements in process B.

How many different values of X are possible after both processes finish executing? Ans = 4 (5, 8, 9, 10)

Ques) $x = 5$

P1	P2	P3
$x = x + 2$	$x = x - 1$	$x = x + 3$

max possible value of x, when all processes completed = 10 ?

min — " — " — 4 ?

Sol^n P1 & P2 read x together & P1 writes last

$x = \cancel{5}\ \cancel{4}\ 7$

$\Rightarrow$ P3 executes

$x = \cancel{7}\ 10$

$x = 20$

P1	P2	P3	P4	P5
$x = x + 3$	$x = x - 2$	$x = x + 4$	$x = x - 8$	$x = x + 1$

max possible value of x = 28 ? $(20 + 3 + 4 + 1 = 28)$

min " " " = 10 ? $(20 - 2 - 8 = 10)$

2 mins Summary

Topic Synchronization

Topic Race Condition

Topic Critical Section

Happy Learning

THANK - YOU